

Roll No.:

Total No. of pages: 2

B.TECH.- FOURTH SEMESTER (Backlog/Improvement)
END-SEMESTER THEORY EXAMINATION,
April-May 2025

Course Code-COMTC13/CAMTC13/CBMTTC13/CDMTTC13/CIMTC13
 Course Title-Probability and Stochastic Processes
 Time: 3:00 hours Max Marks-50

Note: Attempt all the five questions. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q.1.	Attempt any 2 parts of the following		
1(a)	The diameter of an electric cable, say X is assumed to be a continuous random variable. Check whether $f(x) = \begin{cases} 6x(1-x), & \text{for } 0 \leq x \leq 1, \\ 0, & \text{elsewhere.} \end{cases}$ is a pdf. Determine a number c such that $P(X < c) = P(X > c)$.	5	CO1
1(b)	Let X be a random variable having the pdf $f(x) = \begin{cases} \frac{4}{\pi(1+x^2)}, & \text{for } 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$ Find the mathematical expectation and standard deviation for X .	5	CO1
1(c)	The probability mass function of a discrete random variable X is given as $f(x) = \begin{cases} ck^2, & k = 1, 2, 3; \\ c(7-k), & k = 4, 5, 6; \\ 0, & \text{otherwise.} \end{cases}$ (a) Find c . (b) Calculate the second central moment.	5	CO1
Q.2.	Attempt any 2 parts of the following		
2(a)	For a random variable X , use Chebyshev's inequality to calculate (i) $P(5 < X < 15)$ (ii) $P(X - 10 < 3)$ with mean $\mu = 10$ and variance $\sigma^2 = 4$.	5	CO2
2(b)	The average number of phone calls/minute coming into a switch board between 2 and 4 PM is 2.5. Determine the probability that during one particular minute there will be (a) 2, (b) 4 or fewer, (c) more than 6, (d) at most 5, (e) at least 0 calls.	5	CO2
2(c)	Let X be a random variable following Gamma distribution. Find its mean and variance.	5	CO2

	iii) Comment on whether this is a narrowband or wideband FM system.		
3c	Explain the operation of discriminator circuit followed by envelope detector as an FM demodulator. Describe how it tracks the FM signal and draw the relevant waveforms at each stage.	5	CO3
Q 4	Attempt any 2 parts of the following	10	4
4a	What is White noise? Draw the graph for Power Spectral Density (PSD) of White noise. What is the significance of PSD in the analysis of noise signals?	5	CO4
4b	Explain its impact of noise on the performance of FM receivers and how it can be minimized.	5	CO4
4c	Compare the performance of DSB-SC receiver in the presence of channel white noise with respect to baseband communication system.	5	CO4
Q 5	Attempt any 2 parts of the following	10	5
5a	What is aliasing? Explain how it occurs and how it can be avoided.	5	CO5
5b	Explain the working principle of Pulse Amplitude Modulation (PAM). Draw all relevant waveforms.	5	CO5
5c	Draw the block diagram for Pulse Code Modulation (PCM) and explain each block with corresponding signal interpretation. List the advantages and limitations of PCM.	5	CO5

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**B.TECH 4th SEMESTER
END-SEMESTER EXAMINATION, APRIL-MAY, 2025**

Course Title- **COMMUNICATION ENGINEERING**
Course Code- **EAEP405**

Max. Marks- 50

Time- 03 Hours

Note: - Attempt all questions in given order only. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following	10	1
1a	What are the advantages of modulation in communication systems? Name different analog modulation techniques, explain the principle and draw their waveforms in the time domain.	5	CO1
1b	Draw the electromagnetic spectrum and label the major communication frequency bands with one application of each. Briefly explain how frequency bands are allocated for communication.	5	CO1
1c	Define the canonical representation of a band pass signal. How does it help in simplifying system analysis? Explain in detail.	5	CO1
Q 2	Attempt any 2 parts of the following	10	2
2a	In an AM system, the carrier signal peak amplitude is 6 V and 1MHz frequency i) Derive the expression of the AM signal for message signal $m(t) = 2 \cos(2\pi \times 1\text{kHz} \times t)$ ii) Draw the spectrum of the AM wave iii) Compute the percentage of power contained in the sidebands.	5	CO2
2b	With suitable circuit and waveforms, explain the generation of DSB-SC waves using ring modulator. Draw the spectrum and highlight their advantages.	5	CO2
2c	Draw and explain the block diagram of a Superhetrodyne Receiver. How does it help in efficient AM broadcasting and reception? Discuss the role of local oscillator and IF stage.	5	CO2
Q 3	Attempt any 2 parts of the following	10	3
3a	Define Frequency Modulation and Phase Modulation. Derive the relationship between frequency modulation (FM) and phase modulation (PM) with the help of mathematical expressions.	5	CO3
3b	In an FM system, the maximum frequency deviation is 75 kHz and the highest modulating signal frequency is 15 kHz. i) Calculate the modulation index. ii) Determine the approximate bandwidth required for transmission using Carson's Rule.	5	CO3

End Semester of 4th Sem B.Tech.
 END-SEMESTER EXAMINATION, APRIL- MAY, 2025
 COURSE CODE: CBMTC405 & CINTC405
 COURSE TITLE: PROBABILITY THEORY AND STATISTICAL METHODS
 TIME: 3 HRS

Note: - Attempt all five questions. Missing data/ information if any, may be suitably assumed & mentioned in the answer. MAX. MARKS: 50

Q.No.	Attempt any 2 parts of the questions below	Marks	CO																
1(a)	A random variable X has a uniform distribution over $(-3, 3)$. Compute (i) $P(X < 2)$ (ii) $P(X < 2)$ (iii) $P(X - 2 < 2)$ (iv) find k if $P(X > k) = \frac{1}{3}$.	05	CO1																
1(b)	A box contains 4 white, 3 black and 5 red balls. A ball is drawn from the box at random, its colour noted and then the ball is replaced. If 6 balls are drawn in this manner (with replacement) find the probability that (i) 3 are white, 1 black and 2 red (ii) 2 are white and 4 red.	05	CO1																
1(c)	Find the probability that the income tax official will catch 3 income-tax returns with illegitimate deductions if he randomly selects 5 returns from among 12 returns of which 6 contain illegitimate deductions. Also, calculate the probability that the income tax official will catch 2 income tax returns with illegitimate deduction.	05	CO1																
2(a)	Fit a binomial distribution for the following data: <table border="1" style="margin: 10px auto;"> <tr> <td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr> <tr> <td>f</td><td>2</td><td>14</td><td>20</td><td>34</td><td>22</td><td>8</td></tr> </table>	x	0	1	2	3	4	5	f	2	14	20	34	22	8	05	CO2		
x	0	1	2	3	4	5													
f	2	14	20	34	22	8													
2(b)	For the geometric distribution with p.m.f. $f(x) = 2^{-x}$, $x = 1, 2, 3, \dots$. Show that Chebychev's inequality gives $P(X - 2 \leq 2) > \frac{1}{2}$ while the actual probability is $\frac{15}{16}$.	05	CO2																
2(c)	Compute the first four central moments from the following data: <table border="1" style="margin: 10px auto;"> <tr> <td>x</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>30</td><td>35</td></tr> <tr> <td>f</td><td>8</td><td>15</td><td>20</td><td>32</td><td>23</td><td>17</td><td>5</td></tr> </table> Also calculate skewness and kurtosis.	x	5	10	15	20	25	30	35	f	8	15	20	32	23	17	5	05	CO2
x	5	10	15	20	25	30	35												
f	8	15	20	32	23	17	5												
3(a)	Let $\{X_k\}$ be mutually independent and identically distributed random variables with mean μ and finite variance. If $S_n = X_1 + X_2 + \dots + X_n$. Examine if Weak law of large number holds for the sequence $\{S_n\}$.	05	CO3																
3(b)	Find the moment-generating function for the probability distribution given by number of heads appearing when a fair coin is tossed 3 times. Also derive the moment generating function of Gamma distribution.	05	CO3																
3(c)	Using the principle of least squares, fit the curve $y = ae^{bx}$ to the following data: <table border="1" style="margin: 10px auto;"> <tr> <td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr> <tr> <td>y</td><td>7.209</td><td>5.265</td><td>3.846</td><td>2.809</td><td>2.052</td><td>1.499</td></tr> </table>	x	1	2	3	4	5	6	y	7.209	5.265	3.846	2.809	2.052	1.499	05	CO3		
x	1	2	3	4	5	6													
y	7.209	5.265	3.846	2.809	2.052	1.499													

4(a)	A company has the head office at Kolkata and a branch at Delhi. The personnel director wanted to know if the workers at the two places would like the introduction of a new plan of work and a survey was conducted for this purpose. Out of sample of 500 workers at Kolkata, 62% favoured the new plan. At Delhi out of a sample of 400 workers, 41% were against the new plan. Is there any significant difference between the two groups in their attitude towards the new plan at 5% level of significance.	05	CO4																
4(b)	If $x \geq 1$ is the critical region for testing null hypothesis $H_0: \theta = 2$ against the alternative hypothesis $H_1: \theta = 1$ on the basis of $f(x, \theta) = \theta e^{-\theta x}$, $0 \leq x < \infty$. Find the size of the type I error and type II error.	05	CO4																
4(c)	An ambulance service claims that it takes on the average less than 10 minutes to reach its destination in emergency calls. A sample of 36 calls has a mean of 11 minutes and the variance of 16 minutes. Test the claim at 0.05 level of significance.	05	CO4																
5(a)	The theory predicts the proportion of beans in the four groups G_1, G_2, G_3, G_4 should be in the ratio 9:3:3:1. In an experiment with 1600 beans the numbers in the four groups were 882, 313, 287, and 118. Does the experimental result support the theory?	05	CO5																
5(b)	Samples of two types of electric light bulbs were tested for length of the life and following data were obtained: <table><tr><td></td><td>Type I</td><td>Type II</td></tr><tr><td>Sample No</td><td>8</td><td>7</td></tr><tr><td>Sample Means</td><td>1234 hrs</td><td>1036 hrs</td></tr><tr><td>Sample S.D.</td><td>36 hrs</td><td>40 hrs</td></tr></table> Is the difference in the means sufficient to warrant that type I is superior to type II regarding length of life?		Type I	Type II	Sample No	8	7	Sample Means	1234 hrs	1036 hrs	Sample S.D.	36 hrs	40 hrs	05	CO5				
	Type I	Type II																	
Sample No	8	7																	
Sample Means	1234 hrs	1036 hrs																	
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5(c)	Two sample polls of votes for two candidates A and B for a public office are taken, one from among the residents of rural areas. The results are given in the table. Examine whether the nature of the area is related to voting preference in this election. <table><tr><td>Votes for area</td><td>A</td><td>B</td><td>Total</td></tr><tr><td>Rural</td><td>620</td><td>380</td><td>1000</td></tr><tr><td>Urban</td><td>550</td><td>450</td><td>1000</td></tr><tr><td>Total</td><td>1170</td><td>830</td><td>2000</td></tr></table>	Votes for area	A	B	Total	Rural	620	380	1000	Urban	550	450	1000	Total	1170	830	2000	05	CO5
Votes for area	A	B	Total																
Rural	620	380	1000																
Urban	550	450	1000																
Total	1170	830	2000																

Chi-Square Table:

	df = 1	df = 2	df = 3	df = 4	df = 5	df = 6
$\chi^2_{0.05}$	3.841	5.991	7.815	9.488	11.070	12.592

t-distribution table:

	df = 11	df = 12	df = 13	df = 14	df = 15
$t_{0.05}$ (one-tailed)	1.80	1.78	1.77	1.76	1.75
$t_{0.05}$ (two-tailed)	2.20	2.18	2.16	2.15	2.13